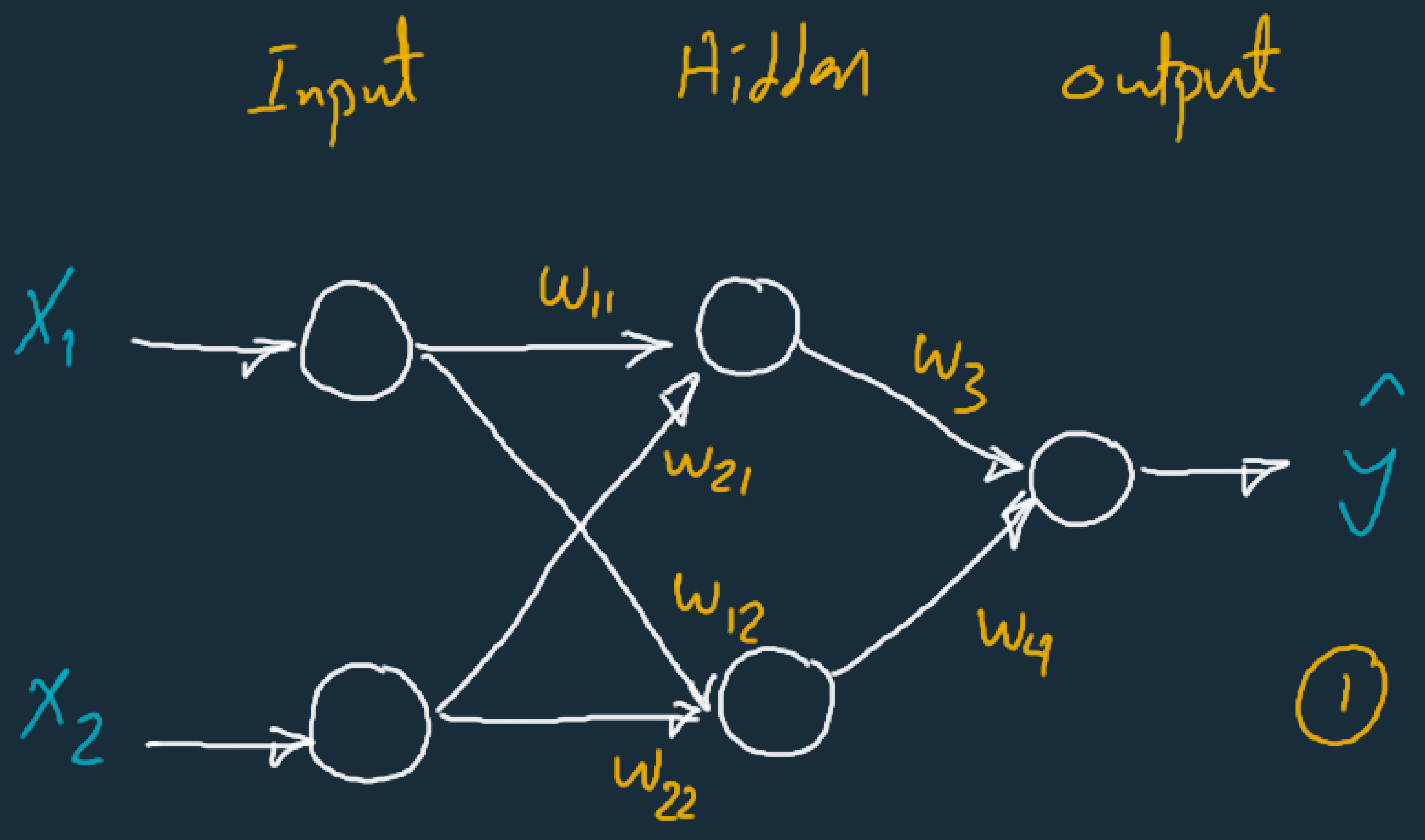


① Forward propagation

$$h_1 = \sigma(w_{11} x_1 + w_{21} x_2)$$

$$h_2 = \sigma(w_{12} x_1 + w_{22} x_2)$$

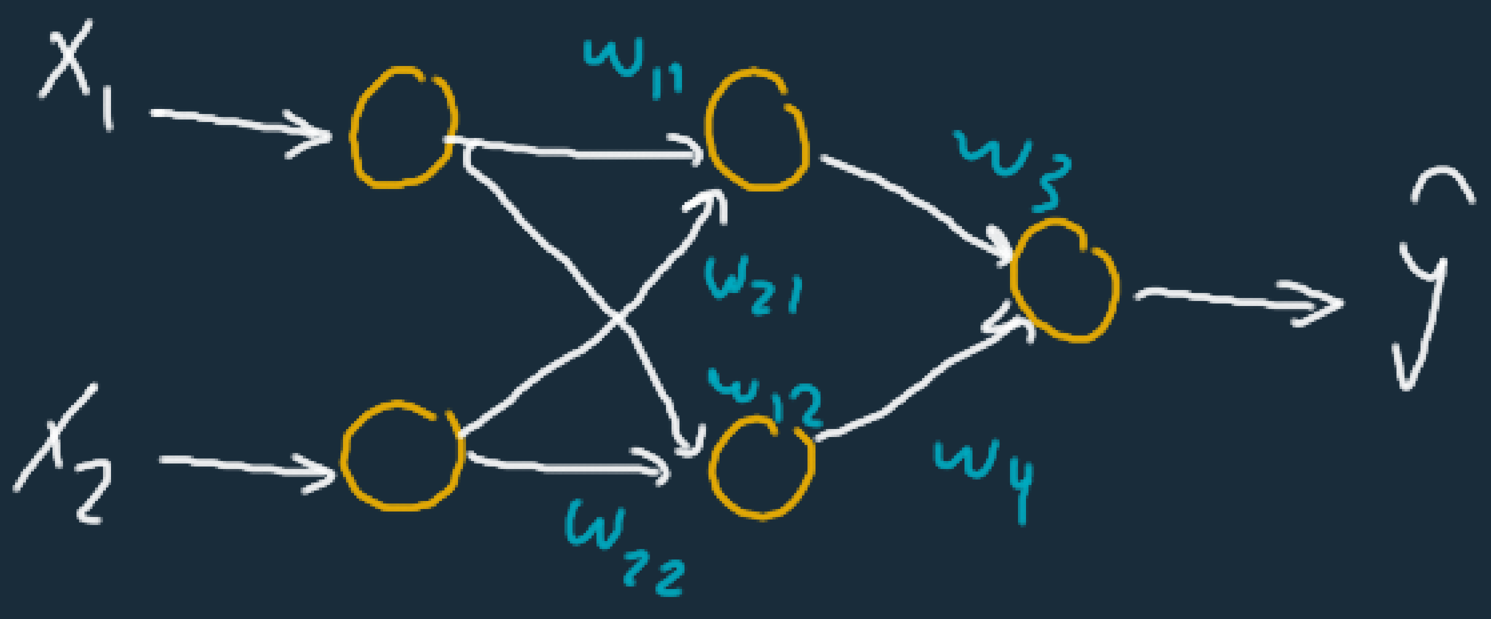
$$\hat{y} = \sigma(w_3 h_1 + w_4 h_2)$$

② Error Estimation

$$J = -y \log \hat{y} - (1-y) \log(1-\hat{y})$$

$$\hookrightarrow f(\underline{\underline{W}})$$

③ Backpropagation



$$\frac{\partial J}{\partial w_3} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_3} = (\hat{y} - y)\, h_1$$

$$w^{new} = w^{old} - \alpha \frac{\partial J}{\partial w}$$

$$\frac{\partial J}{\partial \hat{y}} = -y\left(\frac{1}{\hat{y}}\right) - (1-y)\left(\frac{-1}{1-\hat{y}}\right)$$

$$= \frac{\hat{y}(1-y) - y(1-\hat{y})}{\hat{y}(1-\hat{y})} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$

$$\frac{\partial}{\partial x} \log x = \frac{1}{x}$$

$$\frac{\partial}{\partial x} \sigma(x) = \sigma(x)(1-\sigma(x))$$

$$\frac{\partial \hat{y}}{\partial w_3} = \hat{y}(1-\hat{y})\, h_1$$

$$\frac{\partial J}{\partial w_3} = (\hat{y}-y)h_1$$

$$\frac{\partial J}{\partial w_4} = (\hat{y}-y)h_2$$

$$\frac{\partial J}{\partial w_{11}} = \underbrace{\frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h_1}}_{} \cdot \frac{\partial h_1}{\partial w_{11}}$$

$$(\hat{y}-y)w_3$$

$$\frac{\partial h_1}{\partial w_{11}} = h_1(1-h_1)x_1$$

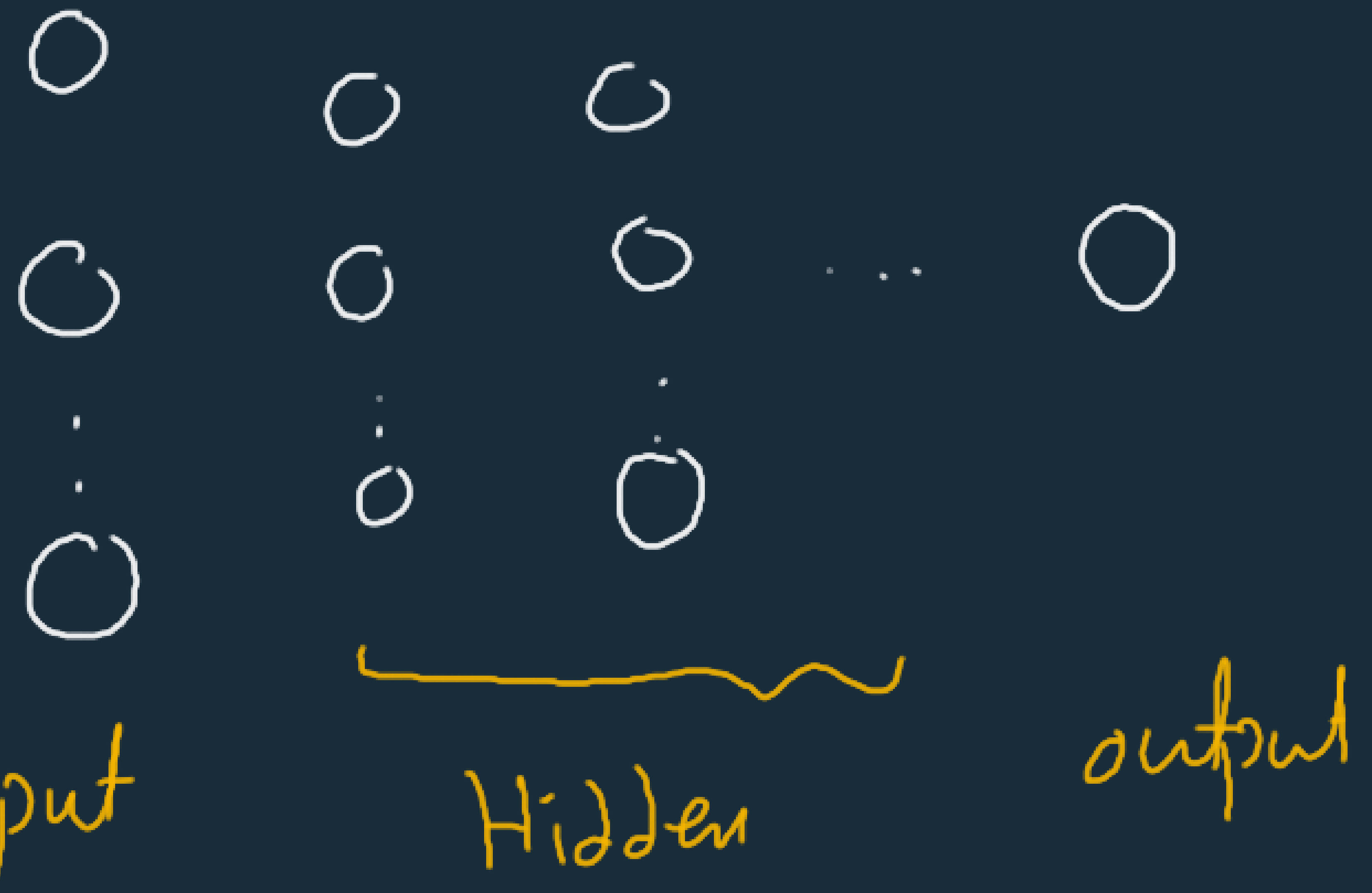
$$\frac{\partial J}{\partial w_{11}} = (\hat{y}-y)w_3 \cdot h_1(1-h_1)x_1$$

$$\frac{\partial J}{\partial w_{12}} = (\hat{y}-y)w_4 \cdot h_2(1-h_2)x_1$$

$$\frac{\partial J}{\partial w_{21}} = (\hat{y}-y)w_3 \cdot h_1(1-h_1)x_2$$

$$\frac{\partial J}{\partial w_{22}} = (\hat{y}-y)w_4 \cdot h_2(1-h_2)X_2$$

Vanishing / Exploding Gradients

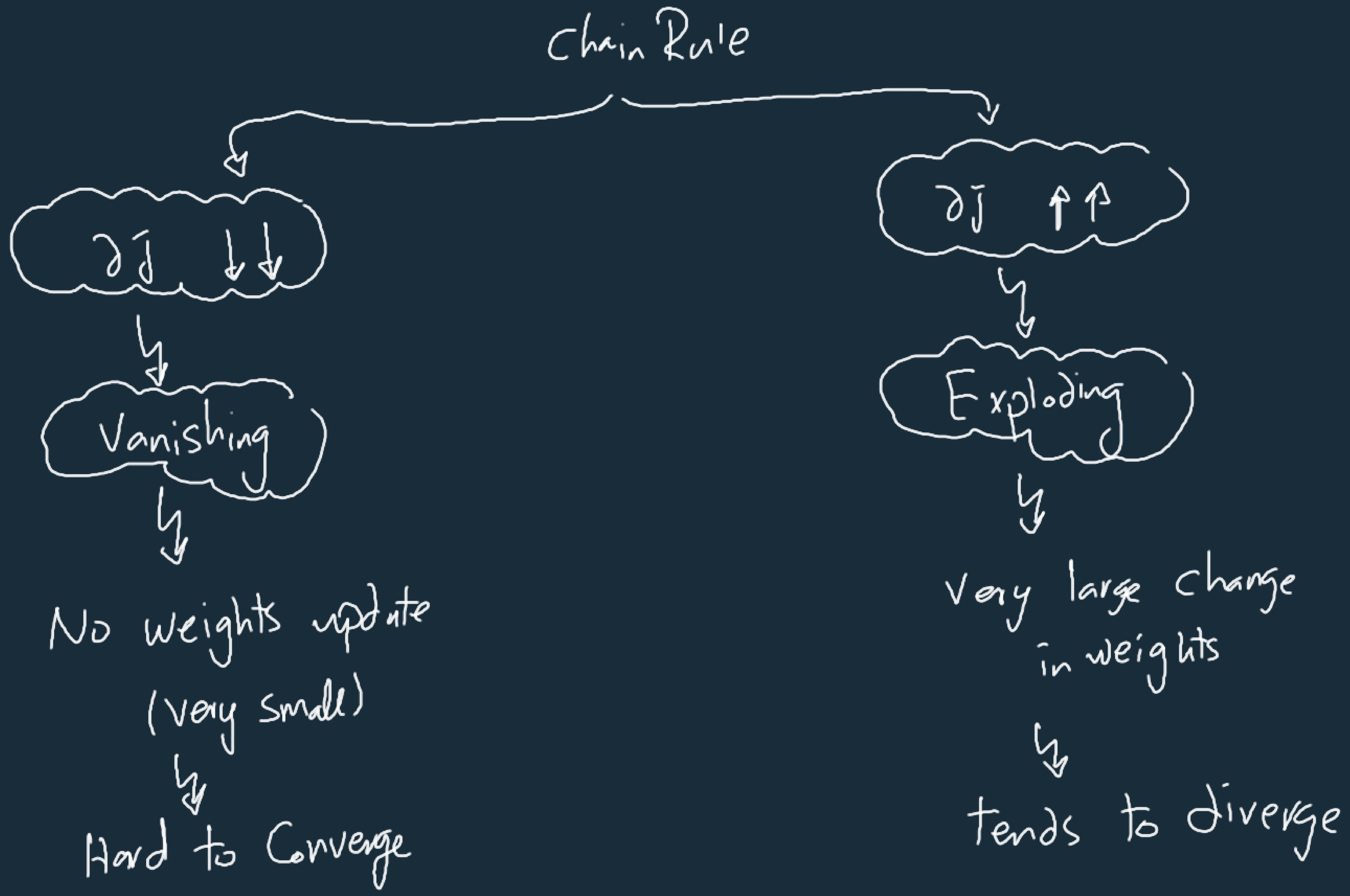


Backpropagation

$$w' = W - \alpha \boxed{\frac{\partial J}{\partial w}}$$

gradient

chain Rule


Chain Rule
∂J ↓↓
Vanishing
No weights update
(very small)
Hard to Converge
∂J ↑↑
Exploding
Very large change
in weights
tends to diverge